

1.3. HOMOGENEOUS EQUATIONS

DEF: A system of equations is called **homogeneous** if all of its constant terms are zero. This is the same as saying that all the equations in the system take the form

$$a_1x_1 + a_2x_2 + \dots + a_{n-1}x_{n-1} + a_nx_n = 0.$$

NOTE: Homogeneous linear systems are always consistent, since $x_1=0, x_2=0, \dots, x_n=0$ is a solution. This is called the **trivial solution**. Any nonzero solution will also be called a **nontrivial solution**.

Because homogeneous linear systems are consistent, they either have exactly one solution (the trivial one) or infinitely many (when there are parameters involved). In other words, if a homogeneous system has a nontrivial solution, then it automatically has infinitely many nontrivial solutions.

THEOREM: If a homogeneous linear system has more variables than equations, then it has infinitely many nontrivial solutions.

PROOF

Let r be the rank of the augmented matrix of the system, m the number of equations in the system and n the number of variables. We have $r \leq m$ (from the definition of rank) and $m < n$ (because there are more variables than equations), so $r < n$. From the theorem at the end of Section 1.2, the general solution involves $n-r > 0$ parameters, so the system has infinitely many (nontrivial) solutions. //

LINEAR COMBINATIONS, BASIC SOLUTIONS

A column matrix will also be called a **vector**. We use the notation $\vec{x}, \vec{y}, \vec{z}, \vec{a}, \vec{b}, \vec{c}, \vec{v}$, etc., to denote vectors. given two vectors (with the same number of entries)

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \text{and} \quad \vec{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix},$$

we say that they are **equal** if $x_1=y_1, x_2=y_2, \dots, x_n=y_n$.

If $\vec{x}, \vec{y}$ are vectors with the same number of entries and k is a number, we define

$$\vec{x} + \vec{y} = \begin{bmatrix} \\ \\ \vdots \\ \end{bmatrix} \quad (\text{Vector sum}) \quad \text{and} \quad k\vec{x} = \begin{bmatrix} \\ \\ \vdots \\ \end{bmatrix} \quad (\text{Scalar product})$$

The zero vector (i.e. the vector with all entries set to zero) is denoted $\vec{0}$.

DEF: Suppose $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_m, \vec{v}$ are vectors (of the same length). We say that $\vec{v}$ is a linear combination of the vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_m$ if there exist constants $a_1, \dots, a_m$ such that

$$\vec{v} = a_1 \vec{v}_1 + a_2 \vec{v}_2 + \dots + a_m \vec{v}_m.$$

EXAMPLE: Consider the vectors:

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} 2 \\ 3 \\ 0 \\ 1 \end{bmatrix}, \quad \vec{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 7 \\ 10 \\ 3 \\ 6 \end{bmatrix}$$

Is $\vec{v}$ a linear combination of $\vec{v}_1, \vec{v}_2, \vec{v}_3$? If so, write $\vec{v}$ as such a combination.

SOL

Given a linear system of equations, we can write its solutions as vectors, i.e., we will say that $\vec{x}$ is a solution if $\vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ and $x_1, \dots, x_n$ is a solution to the system.

PROPOSITION: If $\vec{x}_1, \dots, \vec{x}_m$ are solutions to a homogeneous system of linear equations, then any linear combination of $\vec{x}_1, \dots, \vec{x}_m$ is also a solution to the system.

PROOF: For simplicity, let's consider the case of only two solutions (i.e. $m=2$).

Suppose $\vec{x}, \vec{y}$ are solutions to a given homogeneous system (5), and let $s\vec{x} + t\vec{y}$ be a linear combination of $\vec{x}$ and $\vec{y}$ (where s and t are arbitrary numbers). We may write

$$\vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \quad \vec{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}, \quad \text{so } s\vec{x} + t\vec{y} = \begin{bmatrix} \\ \\ \end{bmatrix}$$

An equation of the homogeneous system (5) is of the form $a_1x_1 + \dots + a_nx_n = 0$. Since $\vec{x}$ and $\vec{y}$ are solutions, we have

Let's check that $s\vec{x} + t\vec{y}$ is also a solution:

Thus, $s\vec{x} + t\vec{y}$ is also a solution to any equation on the system (5), as desired. //

In fact, using Gaussian elimination, we can show the following: for a given homogeneous linear system, its general solution arises as a linear combination of a set of particular solutions:

DEF: Given a homogeneous linear system whose general solution involves s parameters, we say that the vectors $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_s$ are **basic solutions** if any solution to the system is a linear combination of $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_s$.

Basic solutions are found using Gaussian elimination.

EXAMPLE: Find basic solutions to the homogeneous system

$$\begin{cases} 3x_1 + x_2 - x_4 = 0 \\ 2x_1 + 5x_2 + 3x_3 + 4x_4 = 0 \end{cases}$$

SOL

The reduced row-echelon form of the augmented matrix is

$$\left[\begin{array}{cccc|c} 3 & 1 & 0 & -1 & 0 \\ 2 & 5 & 3 & 4 & 0 \end{array} \right] \longrightarrow \left[\begin{array}{cccc|c} 1 & 0 & -3/13 & -9/13 & 0 \\ 0 & 1 & 9/13 & 14/13 & 0 \end{array} \right]$$

with corresponding system

$$\begin{cases} \end{cases} \Rightarrow$$

The general solution is

$$\vec{x} =$$

so a set of basic solutions is

$$\vec{x}_1 = \quad , \quad \vec{x}_2 =$$

We could have also written

$$\vec{x} =$$

so another set of basic solutions is

$$\vec{y}_1 = \begin{bmatrix} \\ \\ \end{bmatrix} , \quad \vec{y}_2 = \begin{bmatrix} \\ \\ \end{bmatrix}$$

In general, a nonzero multiple of a basic solution is also a basic solution. Furthermore, we have:

THEOREM: Every homogeneous linear system admits a set of basic solutions.